

SMALL & LARGE INTESTINE

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Learning Outcome

To understand and discuss

- Imaging modalities
- IBD
- Intestinal obstruction
- Polyps and Tumors
- Imaging in emergency cases:
 - Acute abdomen
 - GI bleeding
 - Abdominal trauma

Imaging Techniques

Barium or contrast examination:

- **Small bowel follow-through** : patient drinks about 200–300 mL of barium and its passage through the small intestine is observed by taking films at regular intervals until the barium reaches the colon, usually takes 2–3 hours, but the transit time is very variable. The indications are suspected inflammatory bowel disease (Crohn's disease), suspected stricture, malabsorption, enterocutaneous fistulae, post small bowel resection, to assess small bowel length, in short bowel syndrome, malrotation
- **Enteroclysis**: barium is injected through the tube followed by water or methyl cellulose to propel the barium through the small bowel, allowing distension of bowel and a double-contrast effect, gives excellent mucosal detail. The disadvantage is that it requires intubation with a nasoduodenal tube,
- **Barium enema**: Firstly, the bowel is prepared, barium is run into the colon under gravity through a tube inserted into the rectum. Air is then blown in to push the barium around the colon with the result that the colon is distended with air and the mucosa coated with barium, giving the '**double-contrast**' effect. **Single contrast** enema is only using barium.



Normal barium follow-through. The small intestine, ascending and transverse colon are filled with barium.

The jejunum in the left side of the abdomen has a much more marked mucosal fold pattern than the ileum

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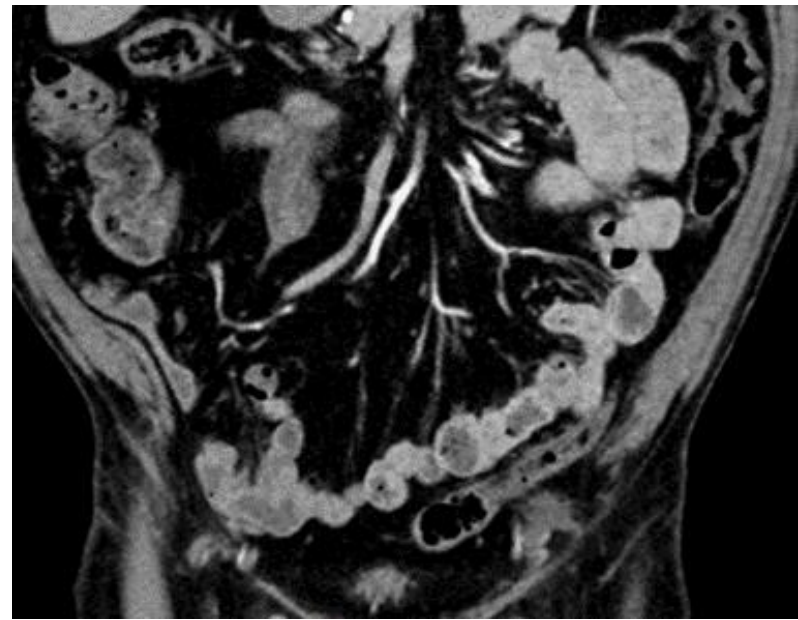


Normal enteroclysis. This technique gives good mucosal detail. The arrow points to the terminal ileum. Note that a tube has been passed through the stomach into the jejunum

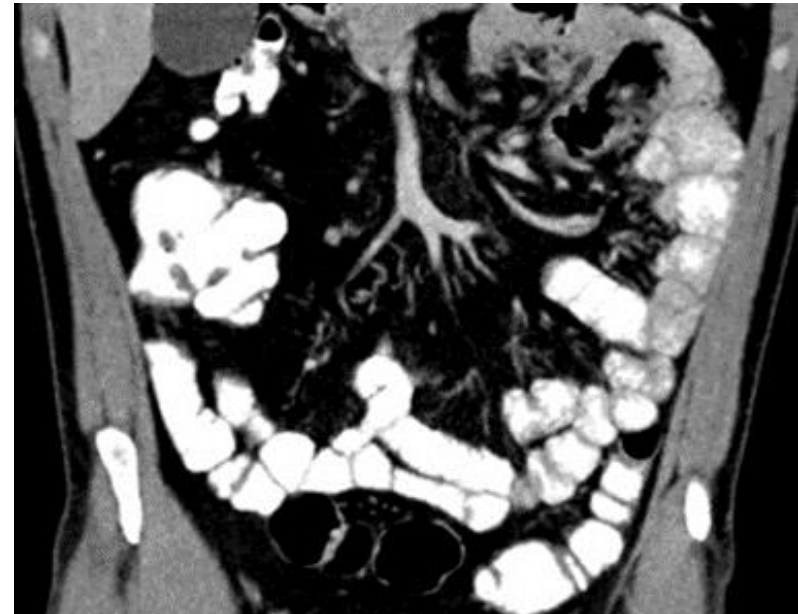
Imaging Techniques

CT: usually performed using IV contrast, sometime using water or Gastrografin, either orally or rectally or both, according to its **indication:**

- Assessment of small and large bowel strictures and masses, also staging them.
- Diagnosis of focal inflammation like diverticulitis, appendicitis, more diffuse inflammation like inflammatory bowel diseases.
- Intestinal obstruction, site of obstruction.
- Mesenteric vessels ischemia, in this case CT angiography is used, which should include arterial and venous phases.
- Intestinal perforation, if erect chest X ray is not diagnostic.
- Help in diagnosis of GI bleeding
- In cases of trauma to exclude intestinal perforation, mesenteric injuries, in addition to diagnosis of solid organ injuries

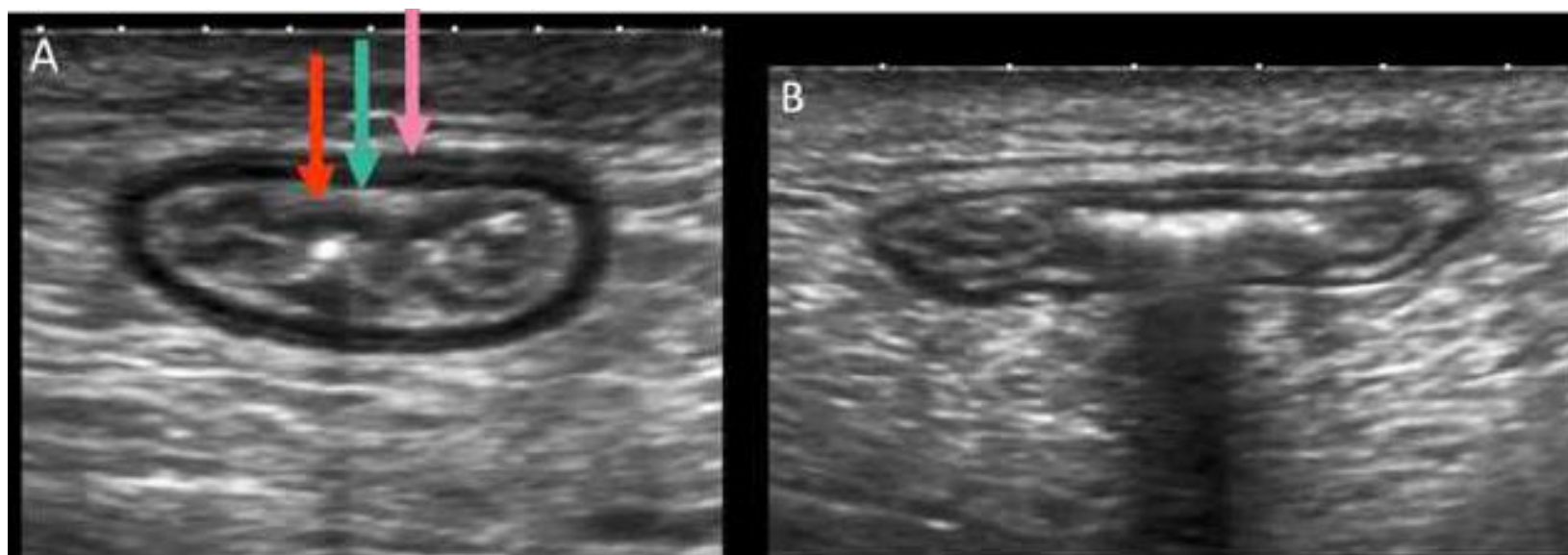


- **Native, with no contrast, when there is absolute contraindication to contrast.**
- **With IV contrast, used in most cases. e.g. obstruction, GI bleeding**
- **IV & Oral contrast, sometimes used to assess stricture, but obscure hemorrhage or enhancement**

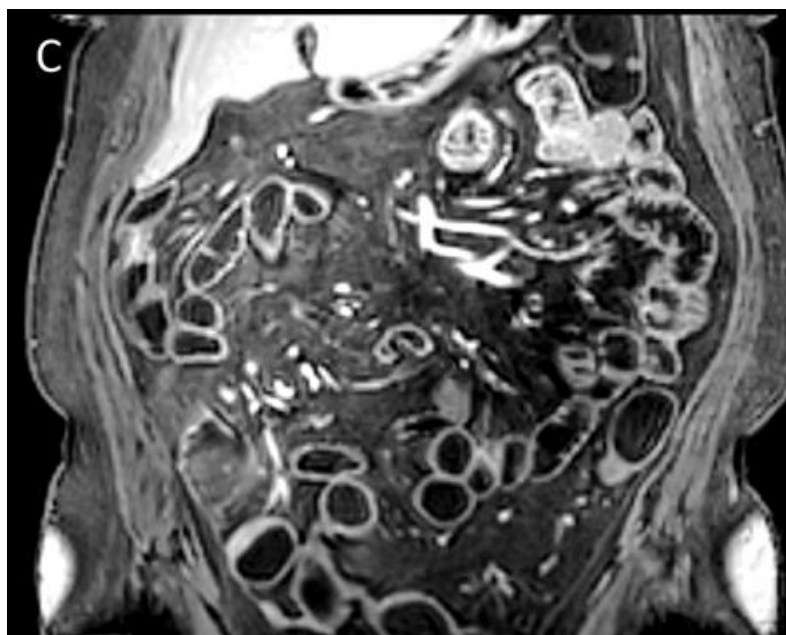


Imaging Techniques

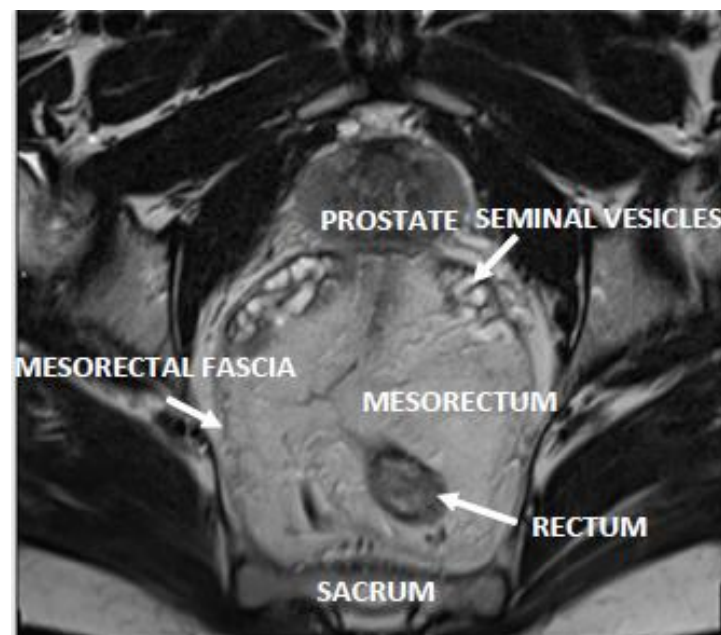
- **Ultrasound:** It can demonstrate bowel thickening and free fluid associated with inflammatory conditions or tumor. It can demonstrate a thickened appendix suggestive of appendicitis, diverticulitis, presence of intussusception.
- **Perineal or transrectal ultrasound** to assess perianal fistula and abscess.
- **MRI:** challenging because movement artifact from peristalsis, but mainly used for rectal and anal imaging where movement is minimal. MR enterography is used to assess inflammatory small intestinal disease like Crohn's disease, but needs preparation and special techniques.
- **Nuclear medicine techniques:** using (MIBG) or octreotide, may be used in cases of suspected neuroendocrine or carcinoid tumours. Technetium-99m (^{99m}Tc) may be used to detect Meckel's diverticulum.



Small intestine US. The red arrow shows the deep mucosa, the turquoise arrow the submucosa & the pink arrow the muscularis propria. Linear echogenic structure in lumen with posterior shadowing is gas.



- **MR enterography**



Rectal MRI

Imaging Findings of disease:

Dilatation

- Small intestine: when diameter is more than 3cm, usually indicates malabsorption, paralytic ileus or small bowel obstruction .
- Large bowel: when diameter more than 6cm is abnormal dilatation. Causes are obstruction, paralytic ileus, volvulus, toxic megacolon in ulcerative colitis, Hirschsprung's disease and megacolon in pediatric age group.
- Colonoscopy and barium enema may only show one end of the stricture, whereas CT will usually demonstrate the cause of the obstruction.
- Paralytic ileus is usually made on clinical grounds with the help of plain films of the abdomen. A water-soluble contrast enema is occasionally undertaken which shows a dilated, but otherwise normal, colon in the case of paralytic ileus .

Imaging Findings of disease:

Narrowing of the lumen more seen in colon, may be due pathology:

- inside the lumen of the bowel (e.g. impacted faeces)
- inside the bowel wall (tumor, may be due to physiological spasm)
- outside the bowel wall due to compression by an extrinsic mass.

When attempting to diagnose the nature of a stricture in the colon the following points should be borne in mind:

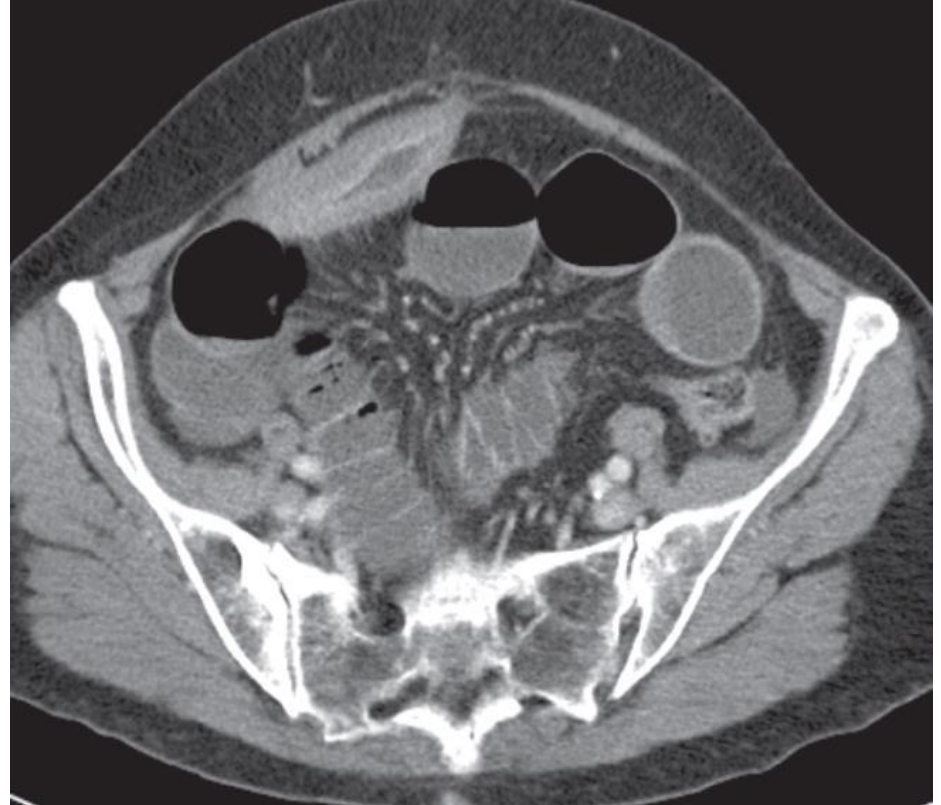
- Shape and length: neoplastic strictures have shouldered edges, an irregular lumen and are rarely more than 6 cm in length. Whereas benign strictures classically have tapered ends, a relatively smooth outline and may be of any length.
- Site of the stricture: diverticular disease are in sigmoid colon. Ischaemic strictures are usually centred somewhere between the splenic flexure and the sigmoid colon. Crohn's disease and tuberculosis have a predilection for the caecum.
- Associated finding: Ulceration, sacculation, diverticuli



- **Dilatation from small bowel obstruction**



Dilatation with short segment stricture in barium enema which caused by carcinoma.



Narrowing. (a) There is a long stricture (arrows) in the ileum due to Crohn's disease and an abnormal mucosal pattern. There is also separation of the abnormal segment from other loops of the bowel. (b) CT in the same patient demonstrating marked thickening of the abnormal loop of small bowel, with a narrowed lumen (white arrows). Several dilated loops of small bowel are also seen (black arrows), due to some obstruction at the level of the stricture.

Inflammatory bowel disease

Ulcerative colitis

- a disease of unknown aetiology characterized by inflammation and ulceration of the colon.
- It involves the rectum, ascends to other parts of colon in continuity.
- When the whole colon is involved, the terminal ileum may become dilated.
- Radiological sign is widespread ulceration which are usually shallow, but in severe cases may be quite deep. There is loss of the normal colonic haustra in the affected portions of the colon.

Inflammatory bowel disease

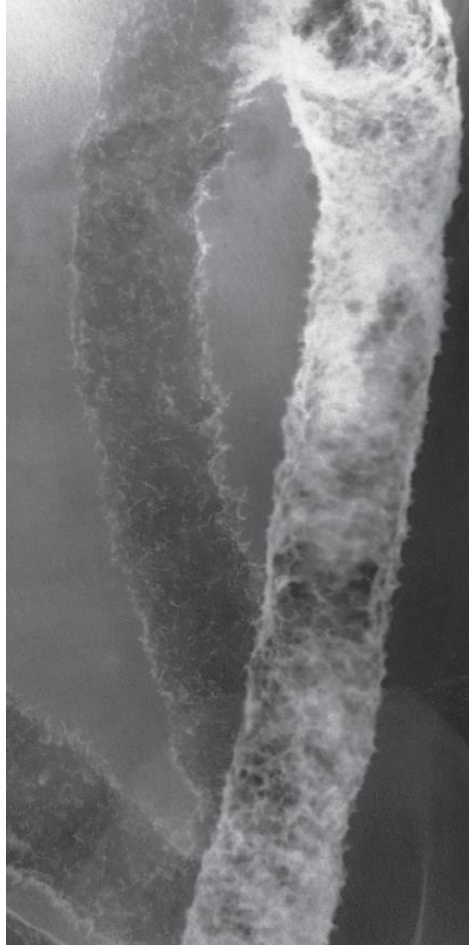
- In severe cases narrowing and shortening of the colon, giving the appearance of a rigid tube, and pseudopolyps are seen. “Pseudopolyps are small filling defects projecting into the lumen of the bowel, formed by swollen mucosa in between the areas of ulceration”
- Strictures are rare, and, when present, are likely to be due to carcinoma.
- On **CT & MRE**, there is bowel wall thickening up to 5–8 mm in continuity from the rectum to the proximal extent of disease. Mucosa enhances brightly “active inflammation”. In chronic disease there is visible fatty infiltration in the submucosa, resulting in a ‘target sign’ appearance.
- **Toxic dilatation (toxic megacolon)** is a serious complication. The diagnosis is made on clinical grounds and on examination of the plain abdominal film or standard CT.

Inflammatory bowel disease

- **Crohn's disease:**
- disease of unknown aetiology characterized by localized areas of chronic granulomatous inflammation, which nearly always affects the **terminal ileum**. It affects several different parts of the small and large intestine and leaving normal intervening bowel, known as **skip lesions**.
- The major signs on barium and cross sectional imaging:
 - **Strictures**, which are extremely variable in length. Sometimes a loop of bowel is so narrow, either from spasm or oedema and fibrosis in the bowel wall, that its appearance has been called the 'string sign'. Dilatation proximal to stricture.
 - **Contraction of the caecum**
 - **Ulcers**; these are sometimes quite deep. Fine ulceration combined with mucosal oedema gives rise to the so-called '**cobblestone**' appearance'.
 - Thickening, distortion or effacement of mucosal folds.
 - **Separation of bowel loops** due to bowel wall thickening or an inflammatory mass in mesentery between loops.
 - **Fistulae** to other small bowel loops, colon, bladder or vagina

Inflammatory bowel disease

- **Ultrasound** may be helpful in identifying thickened loops of bowel, abscess, although these are usually better delineated at CT.
- **CT** shows thick-walled intestinal loops with surrounding fat stranding, prominent mesenteric vessels.
- **MRE** is used recently to demonstrate changes in the wall, and differentiate narrowing due to active inflammation or chronic fibrosis.
- **Perineal MRI** demonstrate detail of perianal fistula, when anal canal is involved.

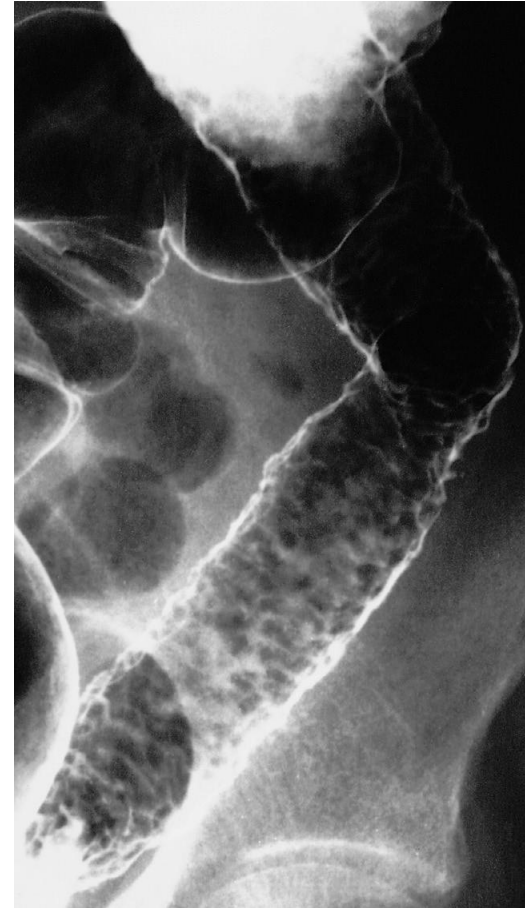


(a) Single contrast. (b) Double contrast. In this case of ulcerative colitis, the ulceration causes the normally smooth outline of the colon to be irregular.

Longstanding UC, haustra are lost, colon becomes narrowed and shortened, resemble a rigid tube. Reflux to ileum through incompetent ileocaecal valve



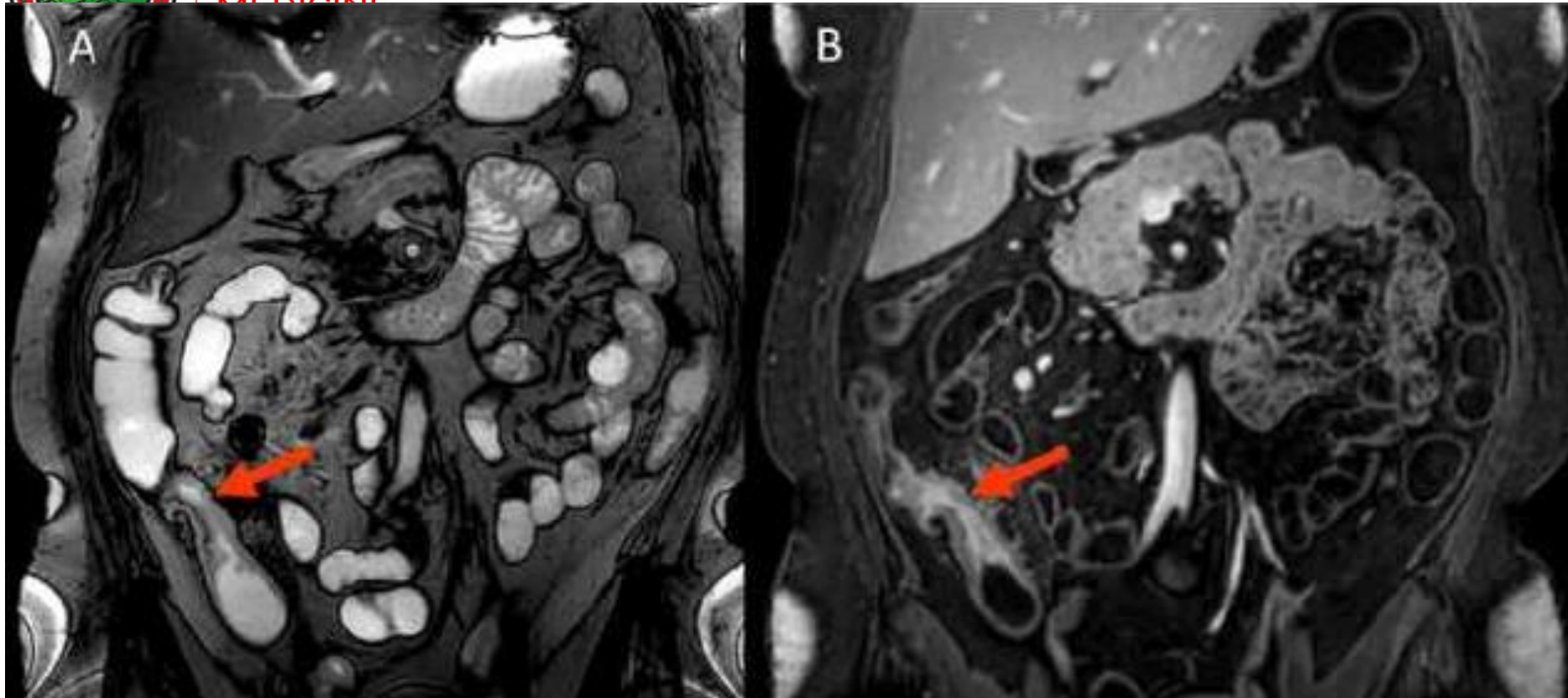
Crohn's disease showing deep ulcers as outward projections



Crohn's disease. The mucosal pattern has a 'cobblestone' appearance due to criss-crossing fine ulceration



Strictures in Crohn's disease. A long stricture is present in the transverse colon (between the curved arrows) and a shorter one in the sigmoid colon (between the small arrows). In this case, the outline of the strictures is irregular due to ulceration. These two abnormal segments with normal intervening bowel are an example of 'skip lesions' – an important diagnostic feature of Crohn's disease.



MRE of Crohn's disease, coronal sequences. The red arrow points at terminal ileum wall thickening and stenosis, the area proximal to the thickened bowel wall is somewhat dilated (pre-stenotic dilation)

Diverticular disease

- Diverticula are sac-like out-pouchings of mucosa through the muscular layer of the bowel wall. Diverticula are common, particularly in the elderly. They are seen in all parts of the colon, but are commonest in the sigmoid colon.
- **Diverticulitis** is when diverticulum is infected, then causes symptoms such as pain, sepsis, diarrhoea or obstruction.
- Barium enema: non-complicated diverticula are seen as spherical out-pouchings with a narrow neck, but when inflamed the neck of diverticulum is occluded.
- CT: in diverticulitis the diverticulum and adjacent bowel wall appears thick-walled with surrounding fat stranding, occasionally perforate, on CT there might be pneumoperitoneum or abscess.
- Stricture from diverticular disease is impossible to differentiate from carcinoma.



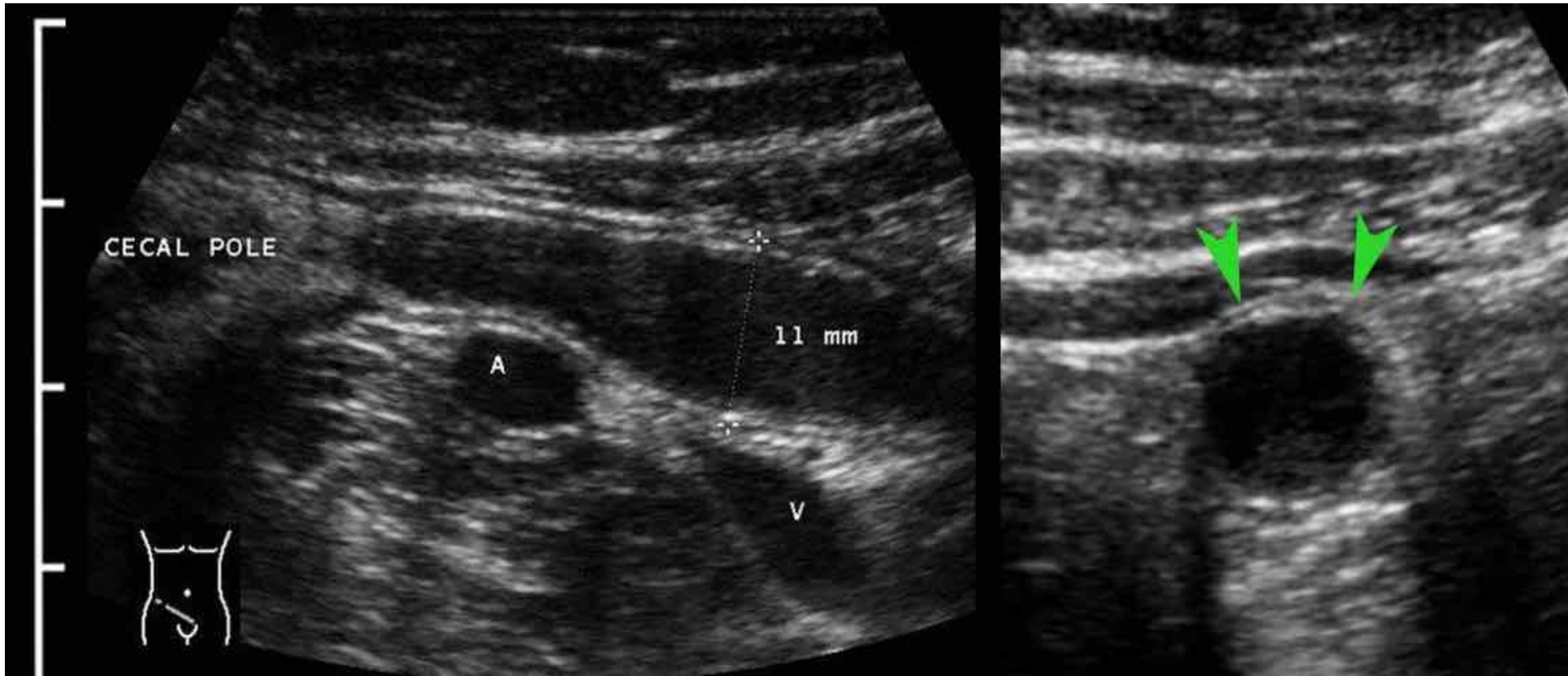
Diverticulosis. A number of diverticula of varying size can be seen arising from the small bowel



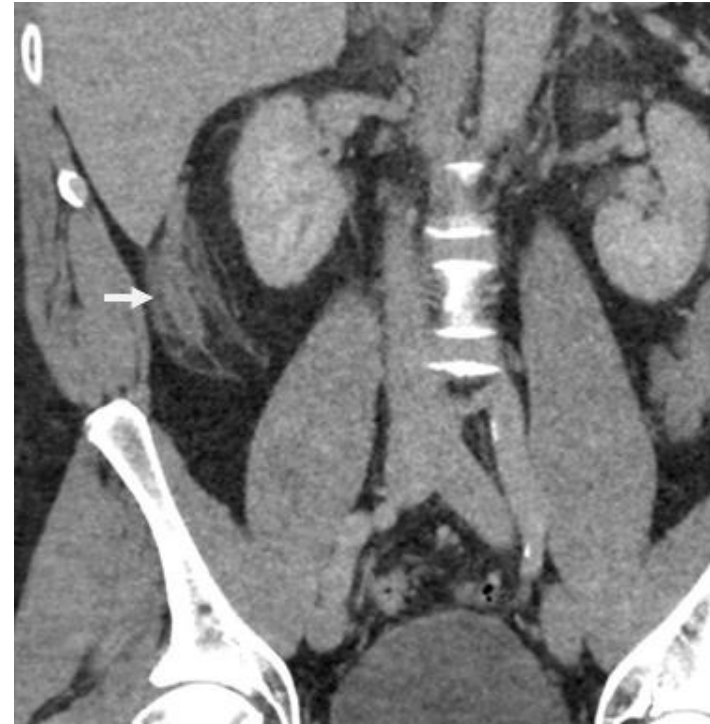
Diverticulitis: fat stranding and inflamed diverticulum

Appendicitis

- In most cases, the diagnosis of appendicitis is obvious clinically and imaging is unnecessary. In cases of doubt the diagnosis can be made with ultrasound, which shows a distended $>6\text{mm}$, non-compressible appendix with a thickened oedematous wall, often accompanied by free fluid and surrounding fat inflammation.
- Similar changes can be demonstrated with CT. An appendix abscess can be diagnosed with CT or ultrasound as a mass in the right iliac fossa.



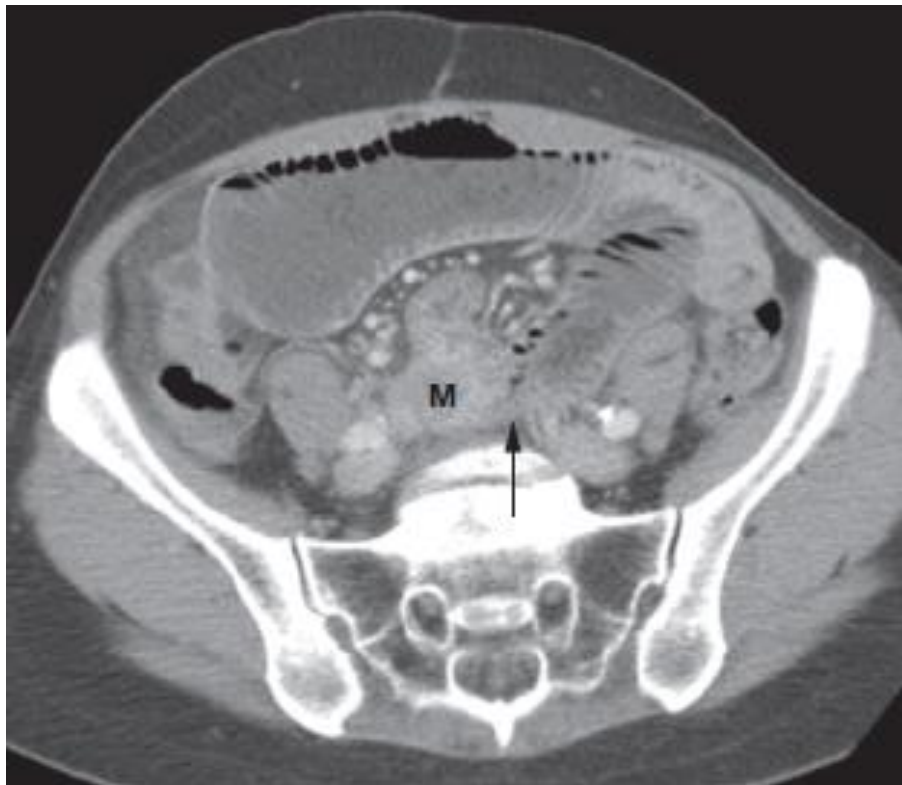
US shows acute appendicitis in longitudinal and short axis, markedly distended blind loop, filled with fluid.



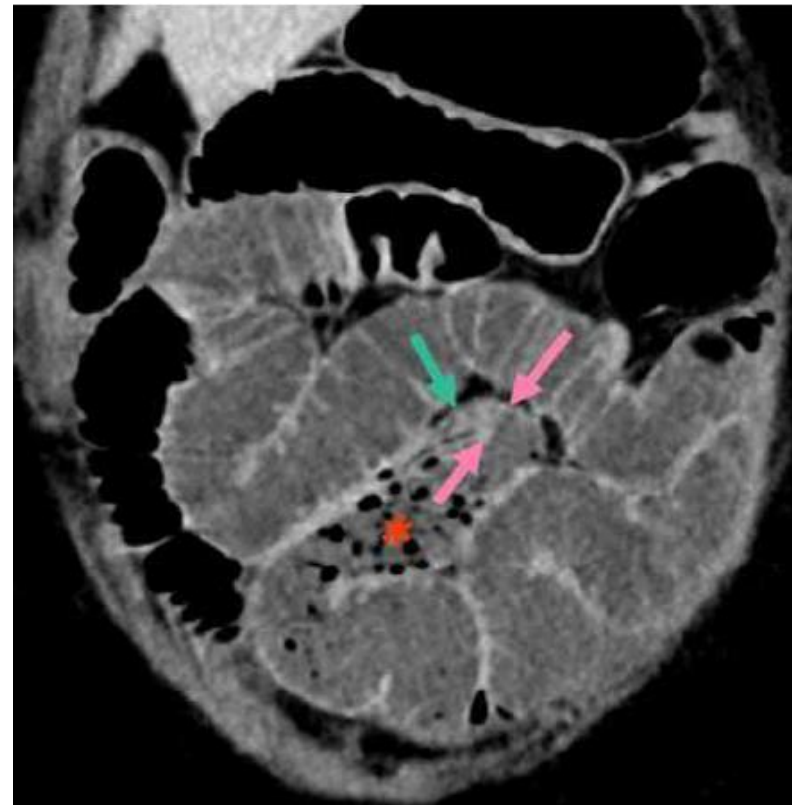
Axial and coronal CT images showing an inflamed appendix (white arrow) with inflammation of the surrounding fat (streaky, reticulated aspect)

Intestinal Obstruction

- Small intestine obstruction mainly due to adhesions or obstructed hernia, but in large intestine mostly due to tumors.
- Diagnosis is usually made on clinical history and examination with the help of plain abdominal films.
- CT is useful technique to demonstrate the site of obstruction by showing the transition zone from dilated to collapsed small bowel, and also to confirm or exclude a mass at the site of obstruction.
- No imaging technique can directly demonstrate adhesions, the commonest cause of small bowel obstruction.



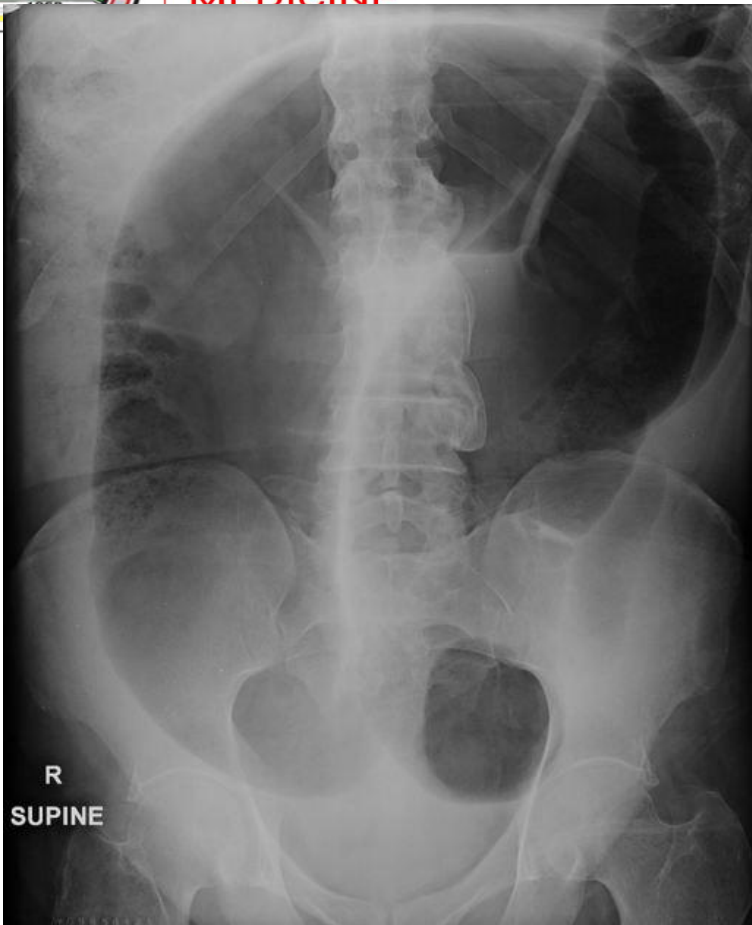
CT of small bowel obstruction in a patient with recurrent ovarian cancer. There is an abrupt change in calibre of the dilated small bowel lumen (arrow) where a mass (M) is infiltrating the bowel wall.



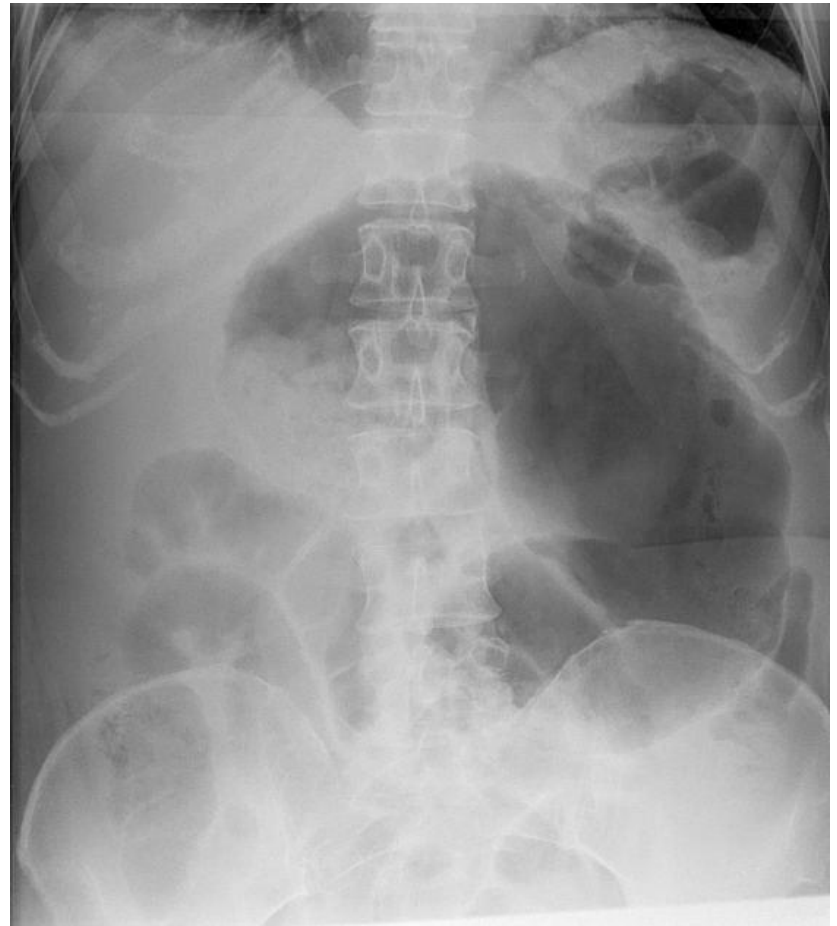
Coronal CT shows small bowel obstruction due to an adhesion, calibre change in the small bowel is seen indicating the adhesion (pink arrows). Proximal of the obstruction there is stasis leading to gas bubbles 'small bowel faeces sign' *, and distally to the obstruction the small bowel loop is collapsed (turquoise arrow).

Volvulus

- A loop of bowel twists on its mesentery. This happens most frequently in the sigmoid colon, particularly when it is redundant, and less often in the caecum. The twisted loop becomes greatly distended and the bowel proximal to the volvulus is obstructed by the twist and may, therefore, also be dilated.
- The diagnosis is usually made on plain abdominal films, but a CT may be helpful in doubtful cases
- **Abdominal X-ray** shows coffee-bean shaped dilated loop of large bowel
- **CT & US** demonstrate “whirl-sign” which denotes twisting of the mesenteric vessels.



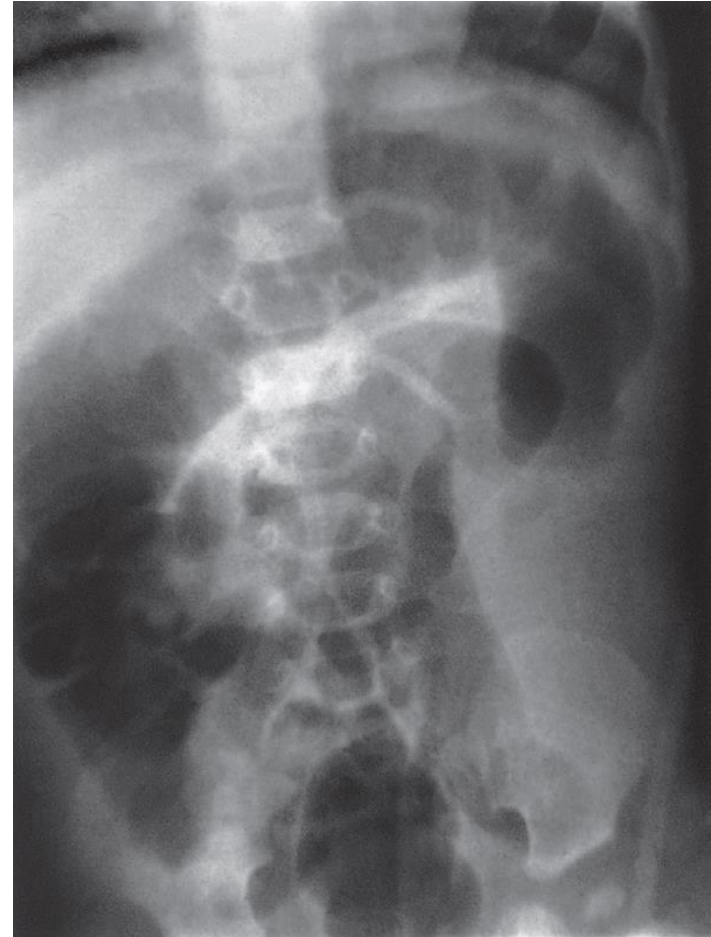
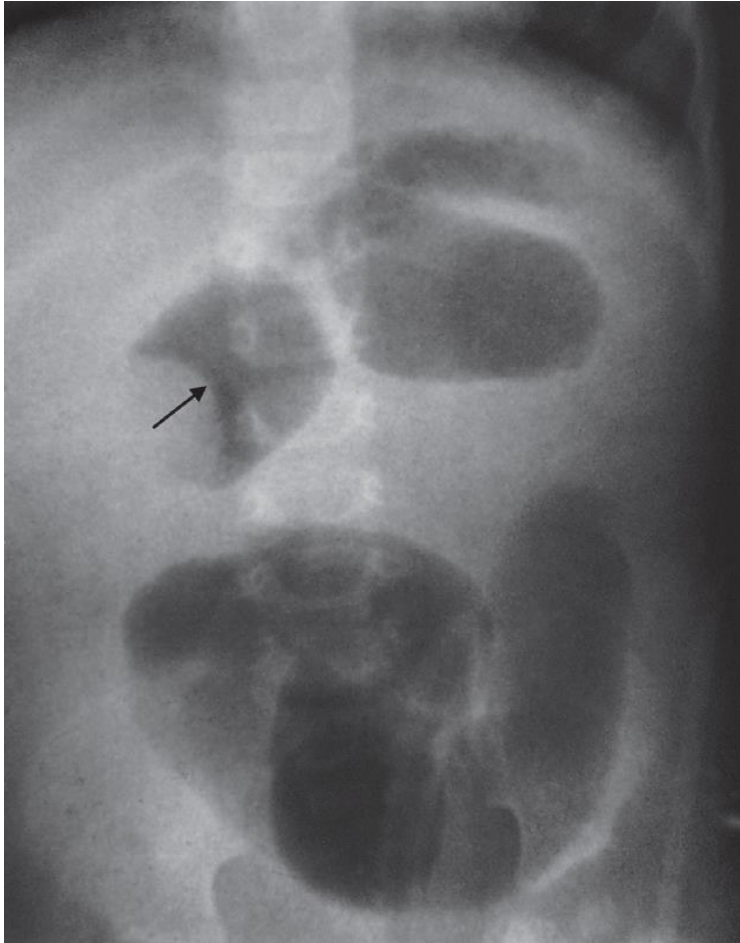
- Sigmoid volvulus



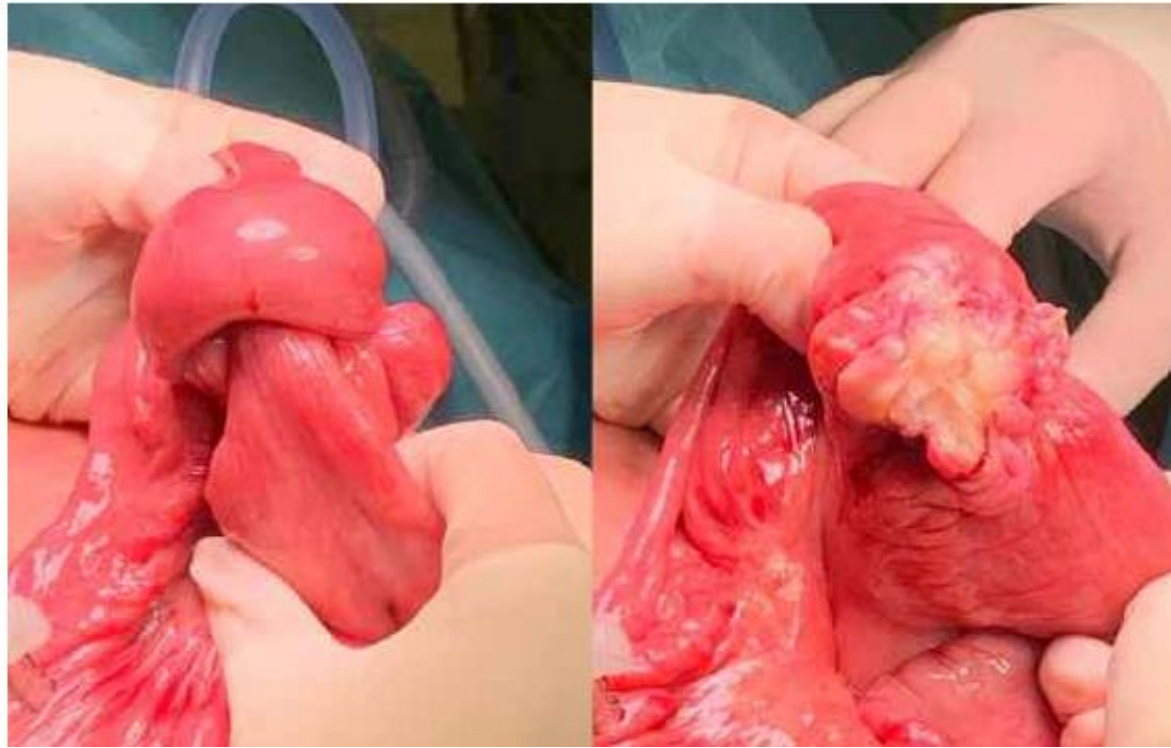
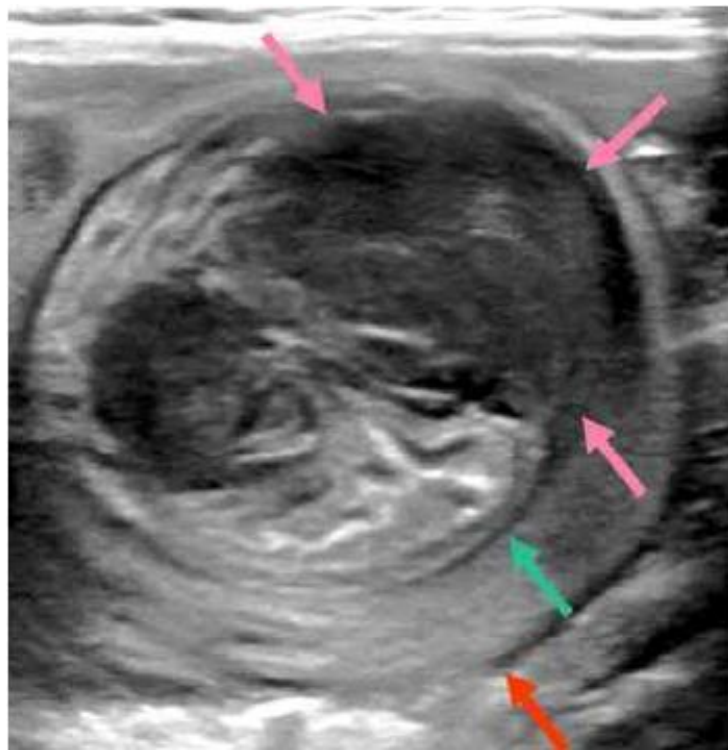
Cecal volvulus

Intussusception

- An intussusception is the invagination of one segment of the bowel into another. Infants and young children are much more liable to intussusception than adults.
- The diagnosis of intussusception **in children** is often made using ultrasound, by diagnosing an abdominal mass often with a hyperechoic centre, often confirmed by an enema with air or carbon dioxide as the contrast agent, and an attempt at reduction of the intussusception is made using the increased pressure of the introduced air or carbon dioxide.
- **In adults**, intussusception is almost always due to the presence of a neoplasm. The diagnosis is usually made on CT. A sausage-shaped mass is demonstrated, which has mesenteric fat within the lumen of the intussusciens.



Intussusception. (a) Film taken during reduction of the intussusception with air insufflated per rectum showing a filling defect in the transverse colon (arrow) owing to ileum invaginated into the colon. (b) Later film showing that the intussusception has been reduced, with air filling the caecum and entering the small bowel.

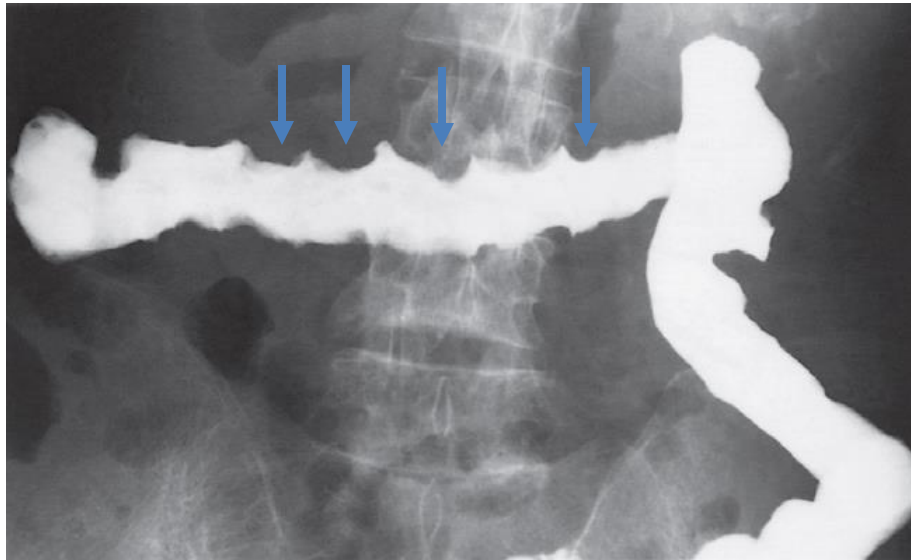


US shows ileo-ileal intussusception due to a tumor “pink arrow”, turquoise arrow points at the ileum with mesentery that is invaginated inside another ileal loop, shown by the red arrow.

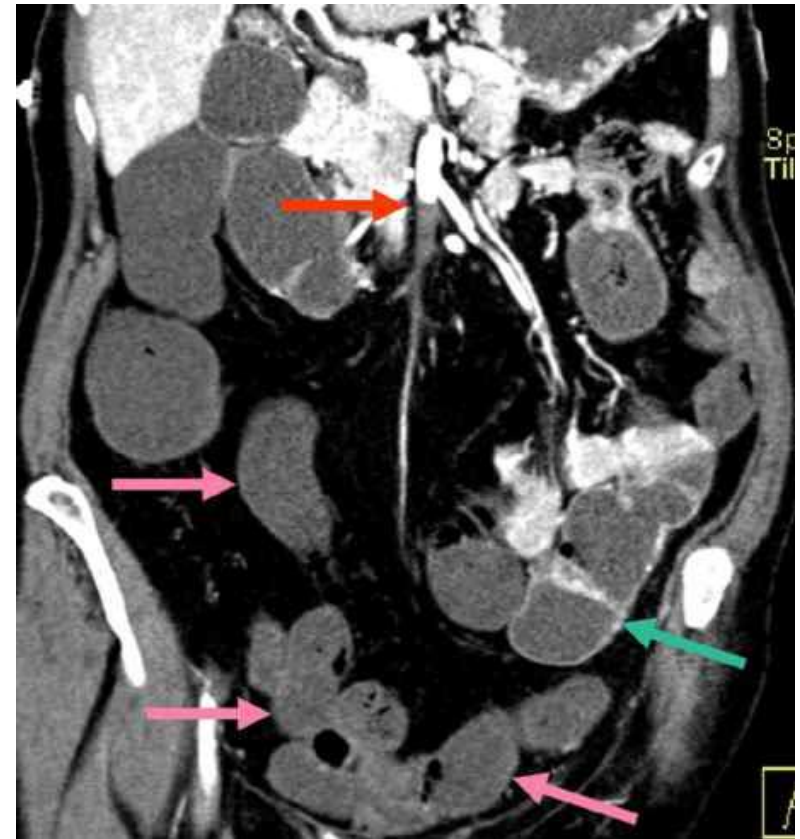
Pictures at surgery with at the left the intussusception and at the right the tumor that had caused the intussusception (lead point) cut open. It was Burkitt lymphoma.

Mesenteric Infarction, Ischaemic colitis

- Small Intestinal infarction, a serious life-threatening condition, is caused by occlusion of the superior mesenteric artery. There may be thickening and oedema of the wall of the small bowel, and gas may be seen within the bowel wall.
- It is diagnosed by **CT**, performed in the arterial and portal venous phase, which shows filling defect “thrombus” in artery or vein, in addition to above mentioned features
- Colon is affected by chronic mesenteric ischaemia, present with vague symptoms, post-prandial pain. On CT there is narrowing of the arteries supplying the bowel, also wall thickening or stricture might be seen.
- On **barium** & occasionally on **plain abdomen X ray**, smooth indentation in lumen called thumb print sign “from mucosal haemorrhage and oedema”, stricture might be seen. Usually centred between the splenic flexure and the sigmoid colon “most vulnerable blood supply”



Thumb-print sign in transverse colon from mucosal edema.



CT angiography shows filling defect in SMA “thrombus or dissection” with small intestinal ischemia “show no enhancement, pink arrows”

Tuberculosis

- It causes contraction of the caecum, dilated terminal ileum.
- On CT, there may be signs of diffuse peritoneal involvement, with ascites, thickening of the omentum, peritoneal and serosal nodules, and lymph node enlargement that may contain necrosis.
- These appearances may be indistinguishable from disseminated intraperitoneal malignancy, or Crohn's disease.
- Biopsy is usually required to establish the diagnosis.



Coronal CT images of TB colitis affecting the right colon (orange arrows). Note the presence of associated enlarged mesenteric nodes (dotted circle)

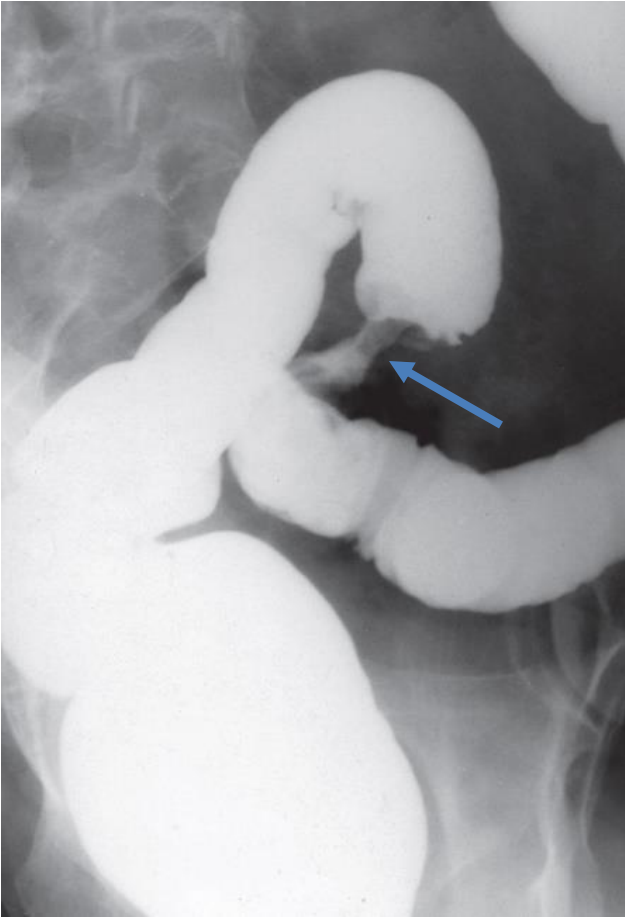
Polyps

- The word ‘polyp’ means a small mass of tissue arising from the wall of the bowel projecting into the lumen. They are best investigated by endoscopy, but may be found at barium enema or CT pneumocolon. It is often impossible on radiological grounds to exclude frank malignancy.
- Only a tiny minority of polyps less than 1 cm in size, and very few less than 2 cm, are cancers.
- The features that suggest **malignancy** are a diameter of more than 2 cm, a short thick stalk or an irregular surface.
- Some polyps might be sessile.

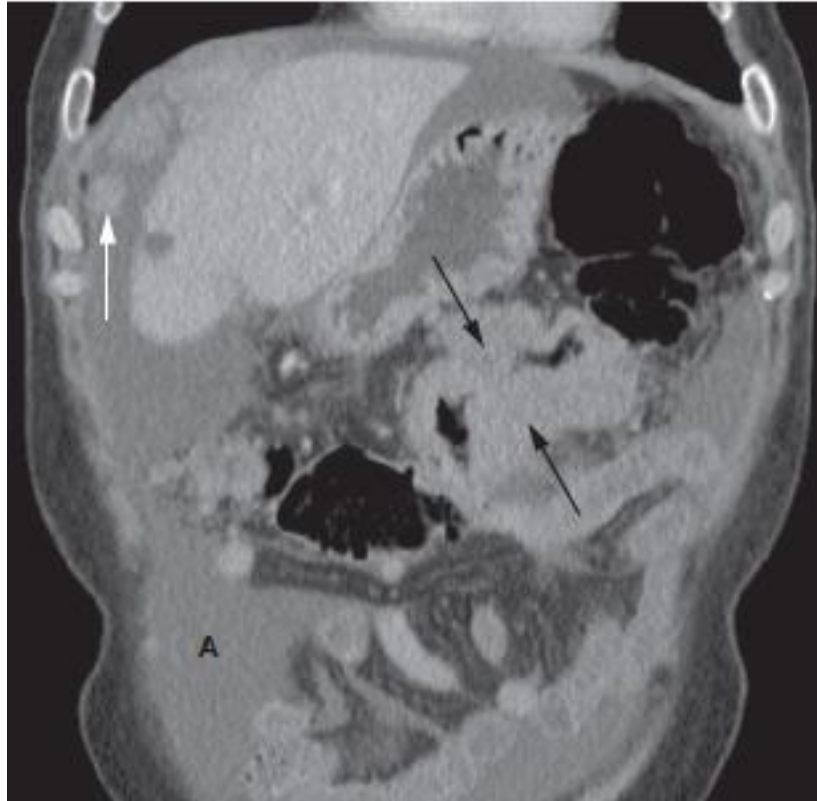
Carcinoma

- Carcinomas may arise anywhere in the colon, uncommon in small intestine, commonest in the rectosigmoid region and caecum.
- The patient with a rectosigmoid carcinoma often has an annular stricture and features of obstruction, whereas with a caecal carcinoma the tumour can become very large without obstructing the bowel, so anaemia and weight loss are the common presenting features.
- **Barium enema:** shows cancer either as irregular stricture with shouldered edges, rarely more than 6 cm in length, or fungating mass as irregular filling defect projecting into the lumen of the bowel.

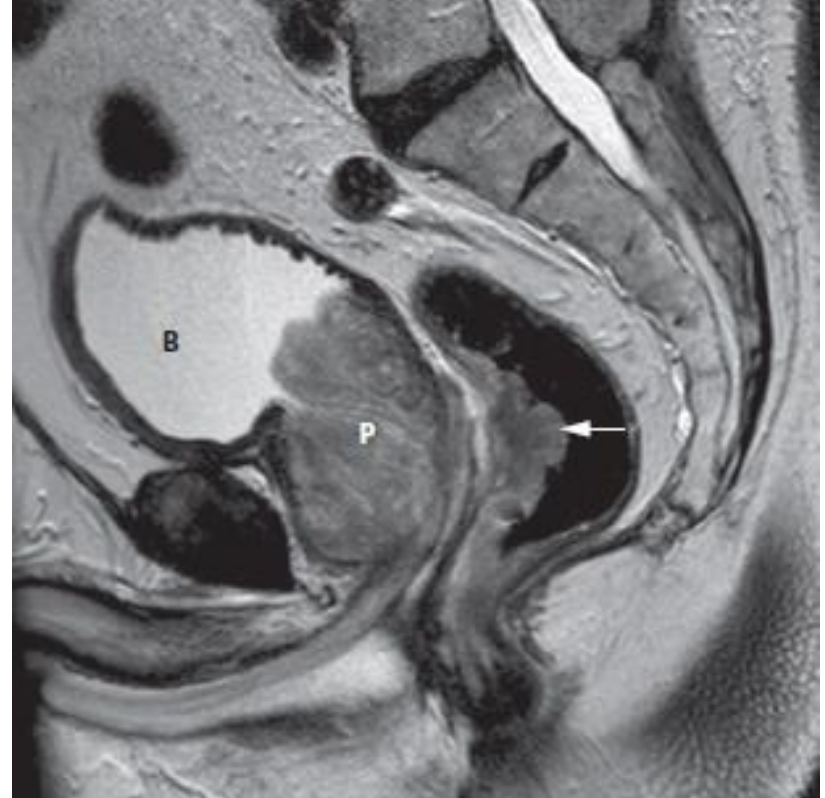
- **CT:** shows carcinoma as thickening of the bowel wall and an irregular narrowing of the lumen of the colon.
- **MRI:** is used for preoperative assessment of the local extent of rectal carcinoma.
- CT is used to demonstrate distant metastases to the liver or lungs and to identify para- aortic nodal disease. CT can also inform on the local staging of a colonic tumor .
- **FDG-PET/CT** is not currently used routinely for initial staging of colorectal cancer, but it is used in patients who are to be considered for potential curative surgery (e.g. hepatic resection for liver metastases) to ensure there are no other sites of metastasis.



Stricture. (a) Barium enema (b) coronal CT showing a short, circumferential narrowing in the sigmoid colon from a carcinoma.



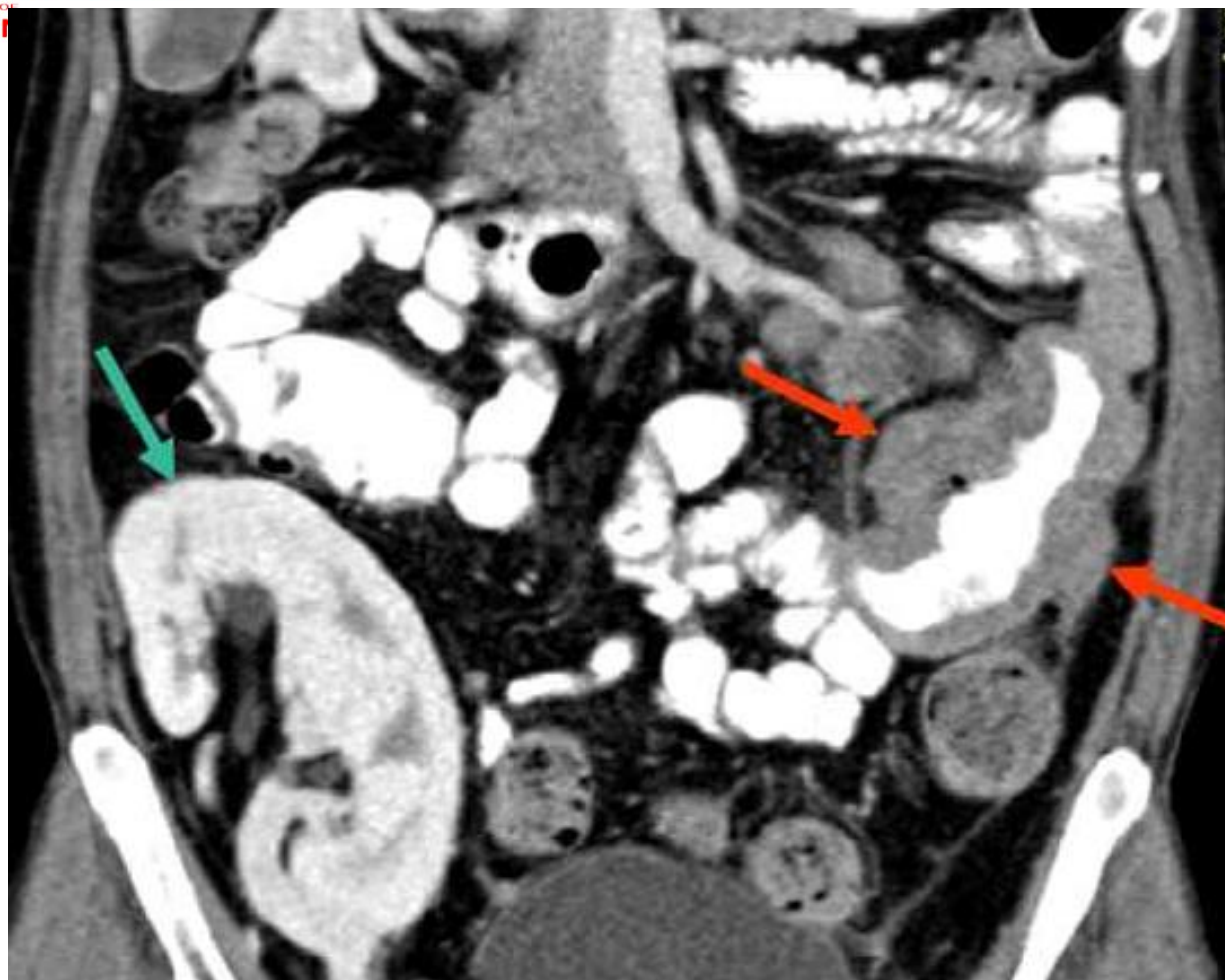
Locally advanced cancer of the transverse colon (black arrows) on a CT coronal reformat. Peritoneal metastases (white arrow) are seen beneath the diaphragm and there is ascites (A) consistent with metastatic disease



MRI Rectal cancer, Sagittal T2-weighted image demonstrating a polypoid growth (arrow) arising from the anterior wall of the rectum.

Lymphoma

- CT and ultrasound can show very marked thickening of the bowel wall, in spite of marked wall thickening there is no narrowing of lumen, which can help differentiate between Crohn's disease and lymphoma.
- CT can also demonstrate mesenteric and para- aortic lymphadenopathy and lymphomatous deposits in the liver and spleen.



Coronal CT, patient with lymphoma (red arrows). Typical of lymphomas is that the tumour despite its size does not obstruct the bowel, the bowel wall is thickened but there is still patent lumen present. Transplanted kidney in RIF turquoise arrow.